

Exotic tori with no action of $SL_n(\mathbb{Z})$

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The obstruction comes from their mapping class groups.

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- (Brown—Rodriguez-Hertz—Wang)
Every infinite action $\mathrm{SL}_n(\mathbb{Z}) \curvearrowright M^{n-1}$ factors through $\mathrm{SL}_n(\mathbb{R})$.

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Σ^3 = Poincare homology sphere

Σ^7 = exotic 7-sphere, generates Θ_7

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Hence, to show $\mathrm{SL}_n(\mathbb{Z})$ does not act on $T^n \# \Sigma$, it suffices to show that the natural map $\mathrm{Diff}^+(T^n \# \Sigma) \rightarrow \mathrm{SL}_n(\mathbb{Z})$ does not split. (Nielsen realization)

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- For $M = T^n \# \Sigma$, $\text{Diff}^+(M) \rightarrow \text{SL}_n(\mathbb{Z})$ is surjective.

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$$\left(\bigoplus_{j=0}^n \Lambda^j(\mathbb{Z}^n) \otimes \Theta_{n-j+1} \right) \oplus (\mathbb{Z}/2\mathbb{Z})^\infty$$

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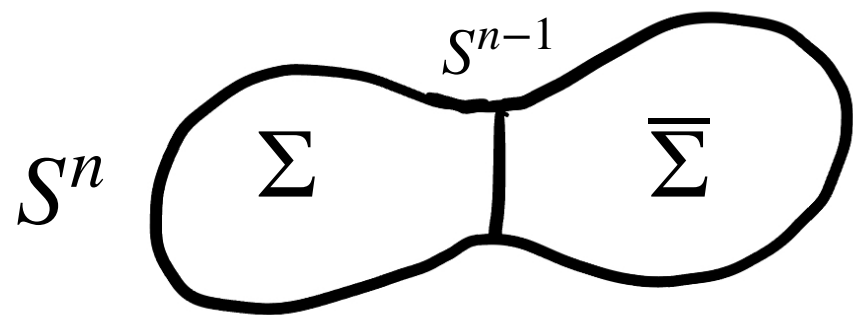
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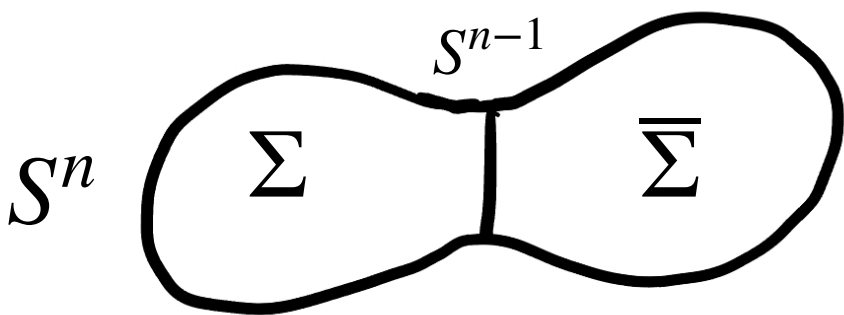
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n	≤ 6 and $\neq 4$	7	8	9	10	11	12	13
Θ_n	0	$\mathbb{Z}/28$	$\mathbb{Z}/2$	$(\mathbb{Z}/2)^{\oplus 2} \oplus \mathbb{Z}/2$	$\mathbb{Z}/6$	$\mathbb{Z}/992$	0	$\mathbb{Z}/3$
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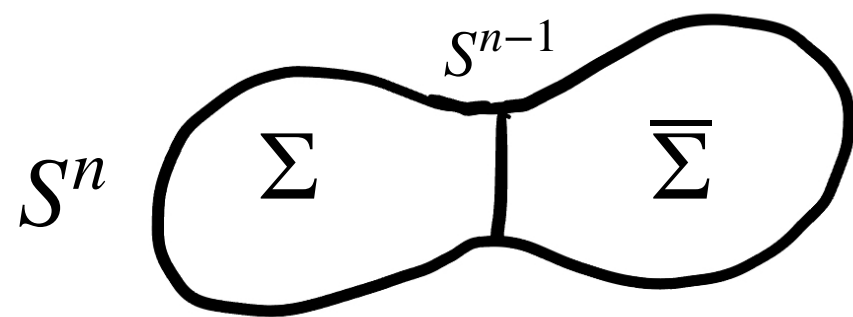
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Example Any homomorphism $\text{SL}_8(\mathbb{Z}) \rightarrow \text{Diff}(T^8 \# \Sigma)$ is trivial.

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Corollary. There exist linear Anosov actions $\mathbb{Z}^2 \curvearrowright T^n$ not realized by any smooth action $\mathbb{Z}^2 \curvearrowright T^n \# \Sigma$.

